Execution-Grounded Hierarchical Graph Learning for Multi-UAV Maritime Data Collection

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***Abstract*—Multi-UAV data collection from maritime sensor buoys couples discrete service assignment with continuous flight control. A high-level decision determines which buoy each unmanned aerial vehicle (UAV) should serve, while the resulting mobility changes link quality, interference, queue service, and the quality of subsequent assignments. This paper presents an execution-grounded hierarchical framework combining a heterogeneous graph neural network (HGNN) assignment scorer with a residual multi-agent proximal policy optimization (MAPPO) controller. A rate-based generator constructs legal joint assignments; the HGNN ranks these assignments using listwise supervision derived from the 15-slot payload obtained by executing each candidate with a frozen low-level controller. The selected assignment is passed to a fixed-duration task interface, and the low-level policy produces per-slot two-dimensional motion residuals around a goal-directed prior, followed by safety projection and bit-queue accounting. In a 6-UAV, 12-buoy simulator, the proposed high-level Scheme C improves mean 48-slot payload over a G1_RATE $k = 6$ baseline by 2.645 Mbit on a 16-scenario development set and by 3.502 Mbit on a separate 16-scenario holdout. The paired-bootstrap 95% confidence intervals are [0.180, 5.017] Mbit and [1.270, 5.395] Mbit, respectively. Low-level optimization attempts have not yet established a reliable gain across seeds. These findings support execution-grounded assignment ranking in the evaluated configuration while exposing the remaining mismatch between short-horizon labels and full-episode performance.**

***Index Terms*—Multi-UAV systems, maritime data collection, heterogeneous graph neural networks, multi-agent reinforcement learning, hierarchical control, learning to rank, trajectory control.**

I. Introduction

Unmanned aerial vehicles (UAVs) can act as mobile data collectors for distributed maritime sensor buoys, providing flexible line-of-sight links where fixed infrastructure is sparse. The control problem is coupled across discrete and continuous decisions: UAV–buoy assignments determine which queues can be served, while UAV movement changes link distances, interference, and the geometry available to later decisions. This coupling makes a single flat policy difficult to train and diagnose, particularly when assignment is combinatorial but motion actions are continuous.

A natural decomposition is hierarchical. A high-level planner periodically selects a joint service assignment, and a low-level controller executes it through motion. This decomposition introduces a coordination problem: an assignment that looks attractive according to an instantaneous rate heuristic may be difficult to execute from current positions, while a motion policy trained under one assignment distribution may not generalize when the high-level policy changes. In addition, a high-level label computed over a short service window need not predict the total payload collected over a longer episode.

This paper develops an execution-grounded hierarchical graph-learning system. The high-level component scores a finite pool of legal joint assignments with a heterogeneous graph encoder and pairwise assignment scorer. Its training labels are generated by running candidate assignments with a frozen low-level controller, rather than relying only on hand-designed instantaneous scores. The selected assignment is wrapped in a fixed-length task option and executed by a residual MAPPO controller. The latter combines a goal-directed motion prior with learned two-dimensional residuals and an explicit safety projector.

The framework is deliberately modular: the current high-level and low-level policies are trained separately and frozen during the formal closed-loop benchmark. It is not an end-to-end jointly optimized hierarchical agent. This distinction matters when interpreting the results and limitations.

The contributions are summarized as follows:

- **Execution-grounded assignment ranking:** high-level decisions are trained through listwise ranking, using the 15-slot payload obtained by executing each legal assignment with the actual frozen low-level controller.
- **Heterogeneous graph scoring under a verified candidate contract:** the scorer encodes UAV nodes, buoy nodes, UAV–buoy edges, UAV–UAV edges, and global context, while legality and score-to-assignment binding are handled explicitly by the generator and inference adapter.
- **A modular closed-loop benchmark:** the high-level ranking change is evaluated with the low-level MAPPO controller fixed, using scenario-level paired comparisons on development validation and a separate holdout set.
- **An empirical diagnosis of hierarchy mismatch:** the experiments distinguish positive full-episode high-level gains from low-level optimization attempts whose gains remain inconclusive, highlighting the limits of short-horizon macro-action returns as proxies for full-episode utility.

II. Related Work

A. UAV Association and Trajectory Control

UAV-enabled wireless networks have been studied through joint optimization of placement, association, resource allocation, and trajectory design. Classical approaches provide

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interpretable constraints and optimization procedures, while deep reinforcement learning has been used to learn trajectory policies under complex communication and mobility dynamics. Representative studies include joint trajectory design in multi-UAV IoT networks and learning-based trajectory optimization with collision avoidance and connectivity constraints [1], [2]. The present work separates joint association from per-slot motion execution and uses the executed controller's short-horizon payload as the high-level ranking target.

B. Graph Learning and Multi-Agent Reinforcement Learning

Graph neural networks provide a natural representation for systems with multiple interacting entity types. MAPPO combines policy-gradient learning with centralized value estimation and is a strong baseline for cooperative multi-agent tasks [3]. The proposed system uses an HGNN to encode UAV–buoy relations, while a residual MAPPO controller handles continuous motion. Unlike a unified graph-MARL policy that jointly samples associations and motion, the current implementation freezes the low-level executor during high-level scorer training.

C. Hierarchical and Execution-Aware Learning

Hierarchical reinforcement learning separates temporally extended decisions from low-level behavior, but the interface can become non-stationary when the lower-level policy changes. Intermediate-target methods such as Waypoint Transformer emphasize structured subgoals in long-horizon behavior [4]. Recent work on queue-aware graph reinforcement learning for UAV-ISAC maritime collection studies a broader joint sensing, association, and motion problem with feasible graph-MARL decisions [5]. This paper instead studies modular assignment ranking where candidates are generated by a deterministic rate-based rule and ranked by measured execution returns under a fixed low-level controller. The contribution is narrower and is not claimed to replace jointly trained association-and-motion methods.

III. System Model and Problem Formulation

A. Maritime Collection Environment

We consider $M = 6$ UAVs and $K = 12$ sensor buoys in a two-dimensional operating region of $160\,\mathrm{m} \times 80\,\mathrm{m}$. The episode horizon is $T = 48$ slots, and the high-level task hold is $H = 15$ slots. The maximum per-slot UAV displacement is $L = 22$ m, and the nominal service radius is $r_s = 16$ m. The environment uses a bit-level queue model with a nominal 1-Mbit block and finite per-buoy queues. The scenario contains two spatial buoy clusters and a simplified cluster-dependent interference model. These are simulator settings, not field-measured maritime parameters.

Let $\mathbf{p}_m(t) \in \mathbb{R}^2$ denote UAV m's position and $\mathbf{b}_k \in \mathbb{R}^2$ denote buoy k's position. The executed normalized motion command $\mathbf{u}_m(t)$ satisfies $\|\mathbf{u}_m(t)\|_2 \le 1$, and the position evolves as

$$\mathbf{p}_m(t+1) = \Pi_\Omega(\mathbf{p}_m(t) + L\mathbf{u}_m(t)), \tag{1}$$

where Π_Ω clips the position to operating region Ω.

B. Association and Communication

At high-level decision time τ, a joint assignment is $\mathbf{a}_\tau = [a_1(\tau), \ldots, a_M(\tau)]$, where $a_m \in \{-1, 0, \ldots, K-1\}$ and $a_m = -1$ denotes an idle UAV. The assignment is a legal partial injection: each UAV has at most one target and a non-idle buoy can be assigned to at most one UAV. The assignment remains fixed for up to $H = 15$ slots, while the low-level controller acts every slot.

The simulator calculates distance-dependent link gains and hypothetical rates from the current geometry. Active service depends on legal association, queue availability, and whether the UAV is within the service radius. Interference is coupled across simultaneous active links, with different weights for same-cluster and cross-cluster links. The exact channel and queue calculations are implemented in the environment and remain identical across the paired high-level benchmark.

Let $q_k(t)$ denote the queue at buoy k, $c_k(t)$ the service capacity offered during slot t, and $d_k(t)$ the actually delivered bits. The queue is served before new arrivals are admitted:

$$d_k(t) = \min\{q_k(t), c_k(t)\}, \tag{2}$$

$$q_k(t+1) = q_k(t) - d_k(t) + a_k^{\mathrm{arr}}(t), \tag{3}$$

where $a_k^{\mathrm{arr}}(t)$ is the accepted arrival amount after queue-capacity clipping. Episode payload is

$$P = \sum_{t=0}^{T-1} \sum_{k=1}^{K} d_k(t). \tag{4}$$

The primary objective is to maximize expected episode payload subject to legal assignment, bounded motion, service-radius, queue, and safety constraints.

C. Hierarchical Decision Interface

At decision time $\tau \in \{0, 15, 30, 45\}$, the high-level planner emits an assignment and packages it into a TaskOption with a fixed hold counter. The low-level controller observes the current environment state and active TaskOption, emits a motion command, and does not rematch targets within the hold window. The next high-level decision uses the evolved state, including updated UAV positions, queues, link quality, and interference.

IV. Execution-Grounded Hierarchical Framework

Fig. 1 summarizes the deployed decision path. The candidate generator, graph scorer, TaskOption interface, low-level controller, and queue environment are distinct modules so that the high-level ranking effect can be measured while the executor remains fixed.

A. Candidate Generation

For each UAV–buoy pair, the generator calculates a current-state hypothetical rate and retains the top-k buoy options for each UAV using a stable tie-breaking rule. The formal configuration uses G1_RATE with $k = 6$. The generator recursively enumerates legal partial assignments: a UAV can be

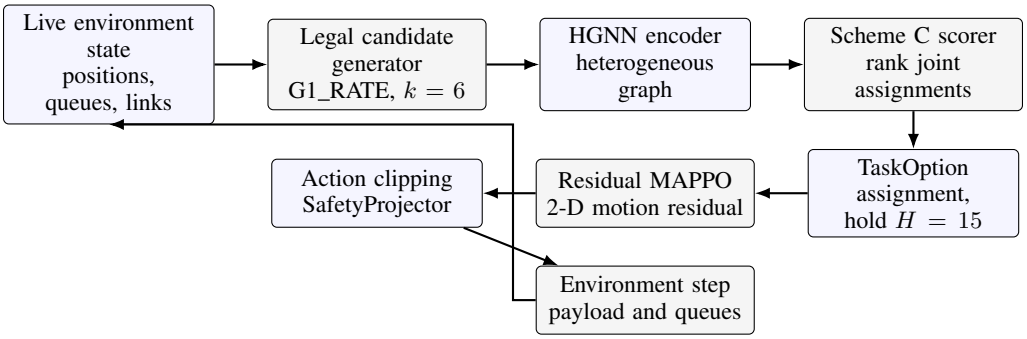


Fig. 1. Closed-loop execution architecture. The high-level assignment is refreshed every 15 slots, whereas the low-level controller acts at every slot.

idle or select one of its retained targets, and duplicate non-idle targets are rejected. In the canonical evaluation configuration, this yields 13,327 legal assignments per decision state. This candidate pool bounds the action space seen by the neural scorer; an assignment absent from the pool cannot be selected by the learned ranker.

B. Heterogeneous Graph Encoder

The high-level input is a heterogeneous graph with UAV nodes, buoy nodes, UAV–buoy edges, UAV–UAV edges, and a global feature vector. The formal tensor shapes are UAV $[1, 6, 8]$, buoy $[1, 12, 8]$, UAV–buoy edge $[1, 6, 12, 4]$, UAV–UAV edge $[1, 6, 6, 2]$, and global $[1, 3]$. The implementation also derives service progress and one-step UAV displacement from the live state. Target labels and future execution outcomes, including J_{15} payload, utility rank, future SINR, and served bits, are excluded from the graph input.

The encoder maps node and edge features to a shared width of 128. Following heterogeneous graph learning principles [6], relation-specific UAV–buoy message passing aggregates pair information into UAV and buoy embeddings; a UAV–UAV layer then aggregates inter-UAV context. Masked aggregation and residual updates preserve node semantics, while global conditioning combines pooled node representations with global context. This produces embeddings used by a matching decoder to score UAV–buoy pairs.

C. Joint Assignment Scoring

Let $\mathbf{h}_m^U$ and $\mathbf{h}_k^B$ denote the encoded UAV and buoy embeddings. A pairwise scorer produces

$$s_{mk} = f_{\text{pair}}\big([\mathbf{h}_m^U, \mathbf{h}_k^B]\big). \tag{5}$$

For candidate assignment $\mathbf{a}_i$, its score is the sum of selected pair scores, with idle assignments contributing zero:

$$S_i = \sum_{m=1}^{M} \mathbb{I}[a_{i,m} \geq 0] s_{m,a_{i,m}}. \tag{6}$$

At inference, the scorer evaluates the legal pool and selects the stable top-scoring row. The chosen score index is explicitly bound to its assignment to avoid candidate-order mismatch. The matching decoder is frozen during Scheme C training; the graph encoder is the principal learned component in this stage.

D. Execution-Grounded Listwise Training

For a state and candidate assignment $\mathbf{a}_i$, a frozen low-level MAPPO controller executes the candidate for 15 slots in a copied environment. The resulting causal payload increment is the label $y_i = J_{15}(s, \mathbf{a}_i)$. The environment is restored between candidate rollouts so candidates are compared from the same initial state. Future arrivals are disabled during label generation, and future outcomes are not supplied to the graph encoder.

The candidate labels are min–max normalized:

$$q_i = \frac{y_i - \min_j y_j}{\max_j y_j - \min_j y_j + \epsilon}. \tag{7}$$

A temperature-scaled target distribution is formed as $p_i = \exp(\beta q_i)/\sum_j \exp(\beta q_j)$ with $\beta = 30$, while the predicted distribution is $\pi_i = \exp(S_i)/\sum_j \exp(S_j)$. Scheme C minimizes the listwise cross-entropy

$$\mathcal{L}_{\text{rank}} = -\sum_i p_i \log \pi_i, \tag{8}$$

with the implementation's fixed target temperature. This listwise objective follows the general learning-to-rank principle of optimizing candidate order rather than absolute scores [7]. It emphasizes relative candidate quality instead of regressing the absolute bit return of each assignment. The ranking target remains a 15-slot execution return, whereas the final system benchmark uses a 48-slot episode return; this horizon mismatch is an explicit limitation.

E. Residual MAPPO Motion Controller

The low-level controller follows an Actor–Critic/MAPPO design. The actor receives self-state, task, communication, and neighboring-UAV observations and outputs a two-dimensional residual. The formal checkpoint is the frozen `phase6g_seed0.pt`; the residual scale is $\alpha = 0.25$. A goal-directed prior $\mathbf{u}_m^{\text{fly}}(t)$ points toward the assigned buoy. If $\boldsymbol{\mu}_m(t)$ is the actor output, the residual is

$$\boldsymbol{\delta}_m(t) = \alpha \tanh(\boldsymbol{\mu}_m(t)), \tag{9}$$

and the desired motion command is

$$\mathbf{u}_m^{\text{des}}(t) = \text{clip}_{\text{unit}}\big(\mathbf{u}_m^{\text{fly}}(t) + \boldsymbol{\delta}_m(t)\big). \tag{10}$$

Algorithm 1 Formal hierarchical episode

Reset the bit-queue environment with the scenario seed.
for $t = 0, \ldots, T-1$ **do**
  if no active TaskOption remains **then**
    Generate the legal G1_RATE $k = 6$ candidate pool.
    Construct the live heterogeneous graph.
    Score candidates with Scheme C and select the stable top row.
    Bind the selected row to a legal assignment and set hold $H = 15$.
  end if
  Build low-level observations under the active TaskOption.
  Compute residual MAPPO actions and apply SafetyProjector.
  Execute the environment step and update queues and payload.
  Decrease the hold counter and continue.
end for
Return the exact cumulative payload and diagnostic metrics.

The joint command is then processed by the SafetyProjector, clipped to the permitted motion range, and applied to the environment. The low-level policy cannot change the assignment during a hold window.

The reported reward is

$$r_t = \frac{\text{served_bits}_t}{10^6} - 0.05 c_{\text{project}}(t) - 0.01 c_{\text{residual}}(t), \quad (11)$$

where the first term rewards actual delivered bits and the remaining terms penalize safety-projection interventions and residual magnitude according to the current implementation. The reward is used for low-level policy optimization; the main system comparison is based on exact episode payload rather than shaped training return.

F. Closed-Loop Execution

Algorithm 1 describes the formal decision loop. Both the high-level scorer and the low-level controller remain frozen during benchmark episodes. The environment and bit-level queue update are shared across the compared high-level policies.

V. Experimental Methodology

A. Scenario Splits and Training

High-level ranking data use scenario IDs 322000–322047 for training, 322048–322063 for development validation, and 322064–322079 as a separate holdout. The holdout is not used for checkpoint selection. Low-level MAPPO training uses a separate scenario namespace (321000–321095), with 321096–321127 used for development evaluation. The formal benchmark loads the frozen Scheme C Epoch 1 scorer and the frozen MAPPO-V2 seed-0 controller; it does not train either module during evaluation.

The low-level training configuration recorded in the repository uses PPO/MAPPO with $\gamma = 0.99$, GAE parameter $\lambda = 0.95$, learning rate 3×10^{-4}, gradient clipping, and multiple random seeds. The high-level Scheme C model starts from an HGNN checkpoint, freezes the matching decoder, and updates the encoder using the listwise objective in (8). The selected high-level checkpoint is Epoch 1 of Scheme C; the formal low-level checkpoint is `phase6g_seed0.pt`.

TABLE I
Canonical Environment and Communication Parameters

Parameter	Value
Number of UAVs, M	6
Number of buoys, K	12
Operating area	160×80 m^2
Episode horizon, T	48 slots
High-level hold, H	15 slots
Maximum displacement	22 m/slot
Service radius	16 m
Nominal block payload	1 Mbit
Bandwidth	1 MHz
Path-loss exponent	2.2
Same-cluster interference factor	1.0
Cross-cluster interference factor	0.08

B. Baselines and Metrics

The primary baseline is G1_RATE $k = 6$. It uses the same legal candidate pool and the same frozen low-level controller as Scheme C, but ranks assignments by the sum of normalized hypothetical pair rates. This paired design isolates the high-level ranking rule. Low-level diagnostics compare MAPPO with a direct `fly_toward` prior, but these diagnostics are not used to claim a stable low-level improvement.

The primary metric is exact cumulative episode payload in bits. Secondary metrics include completed block equivalents, residual queue, average SINR, average interference, service steps, SafetyProjector interventions, and runtime. Statistical comparisons are computed over scenario-level paired differences; time slots are not treated as independent samples. Confidence intervals use 10,000 paired bootstrap resamples with a fixed seed. The independent holdout is used only for the final reported comparison.

VI. Results

A. Offline Candidate-Ranking Performance

On the 16-state development validation set, Scheme C improves ranking metrics relative to the HGNN Epoch 6 initialization. The mean selected J_{15} increases from 18.892 Mbit to 19.240 Mbit. Top-1 accuracy rises from 0.5625 to 0.6667, Top-3 from 0.6250 to 0.7083, and Top-5 from 0.6667 to 0.7083. Normalized regret decreases from 0.0507 to 0.0355. The median paired J_{15} difference remains zero, indicating that the mean improvement is concentrated in a subset of states rather than uniformly distributed.

B. Full-Episode Development and Holdout Results

Table III reports the paired full-episode comparison. On development validation, Scheme C achieves 68.918 Mbit mean payload compared with 66.273 Mbit for G1_RATE $k = 6$,

TABLE II
OFFLINE HIGH-LEVEL RANKING RESULTS

Metric	HGNN Epoch 6	Scheme C Epoch 1
Mean selected J_{15}	18.892 Mbit	19.240 Mbit
Top-1	0.5625	0.6667
Top-3	0.6250	0.7083
Top-5	0.6667	0.7083
Normalized regret	0.0507	0.0355

TABLE III
FULL-EPISODE PAIRED BENCHMARK AGAINST G1_RATE $k = 6$

Metric	Validation	Holdout
Scenarios	16	16
Scheme C mean payload	68.918 Mbit	69.151 Mbit
G1_RATE mean payload	66.273 Mbit	65.648 Mbit
Mean paired gain	2.645 Mbit	3.502 Mbit
95% paired CI	[0.180, 5.017]	[1.270, 5.395]
Positive / tie / negative	11 / 0 / 5	14 / 0 / 2

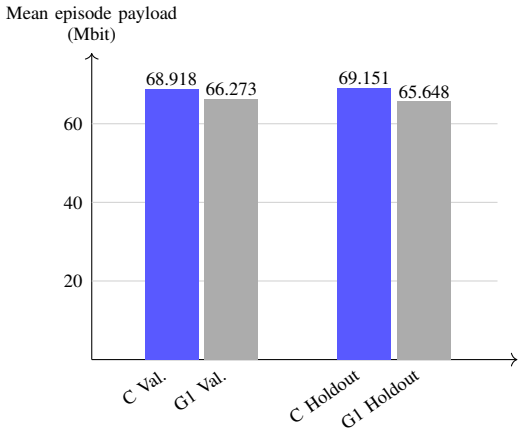


Fig. 2. Mean 48-slot episode payload on development validation and the independent holdout. Values are reproduced from the saved benchmark report.

yielding a mean paired gain of 2.645 Mbit with a 95% confidence interval of [0.180, 5.017] Mbit. Eleven scenarios are positive and five are negative. On the separate 16-scenario holdout, Scheme C achieves 69.151 Mbit compared with 65.648 Mbit, with a mean paired gain of 3.502 Mbit and a 95% interval of [1.270, 5.395] Mbit. Fourteen holdout scenarios are positive and two are negative.

The results support a positive mean gain for the learned high-level ranker in the evaluated configuration, but do not imply that it wins on every scenario. The holdout contains two notable negative cases, demonstrating that a high-quality local assignment can still lead to a less favorable future state. The discrepancy is consistent with the difference between the 15-slot training target and the 48-slot evaluation objective.

C. Runtime and Low-Level Diagnostics

In recorded CPU runs, neural scoring takes approximately 6–10 ms per high-level decision, whereas candidate-pool generation takes approximately 0.18–0.24 s. Thus, candidate construction, not HGNN inference, dominates high-level decision latency. Future work should reduce this cost only if candidate legality and coverage are preserved.

Low-level experiments tested direct two-dimensional velocity control, increased residual scales, distance-potential reward shaping, and training on assignments sampled from Scheme C. None has established a robust payload gain across seeds, so the formal controller remains the frozen residual MAPPO-V2 checkpoint. A matched-state, 15-slot diagnostic reported positive local gains for MAPPO over `fly_toward`, but full-episode results were not consistently positive. This reinforces that short-horizon controller diagnostics should not be substituted for closed-loop episode evaluation.

D. Visual Summary of the Benchmark

VII. DISCUSSION

A. Interpretation of the Current Evidence

The primary result is a positive paired mean payload difference for Scheme C against G1_RATE $k = 6$ when both use the same frozen low-level controller. The design makes the comparison interpretable: the change is the high-level assignment ranking, rather than simultaneous changes to the motion policy, queue accounting, or environment. The independent holdout provides evidence beyond checkpoint selection, although its 16 scenarios remain a limited sample from a single simulator family.

The evidence does not establish that the complete hierarchy is optimal or that the low-level controller is superior to the goal-directed baseline in every setting. Low-level ablations remain inconclusive, and the high-level ranker is restricted to a rate-generated candidate pool. Moreover, a 15-slot execution label may be misaligned with the full 48-slot objective. These issues motivate a staged rather than immediate end-to-end optimization strategy.

B. Limitations

First, the environment is synthetic and two-dimensional, with a fixed set of six UAVs and twelve buoys, simplified distance-based propagation, a two-cluster interference model, and prescribed arrival behavior. The independent holdout is not field-data validation. Second, the high-level candidate pool restricts the reachable assignment set; the learned scorer cannot select candidates that the generator omits. Third, high-level training uses a 15-slot return, while final evaluation spans 48 slots. Fourth, high-level and low-level modules are trained separately and frozen during formal evaluation, so the method does not demonstrate end-to-end joint learning. Finally, low-level residual control is constrained by a `fly_toward` prior and cannot change assignments during a hold window.

C. Future Work

The next steps are to (i) train the low-level controller over a mixture of legal task distributions generated by multiple high-level policies, (ii) audit temporal credit assignment across the 15-slot decision boundaries and 48-slot episode, (iii) reduce candidate-generation cost without sacrificing legality or high-value candidate coverage, and (iv) evaluate additional baselines under a common compute and scenario budget. Only

after module-level stability is established should alternating or joint high–low fine-tuning be evaluated. Robustness to drifting buoys, measured channel traces, energy constraints, and obstacle geometry is also required before broader generalization claims can be made.

VIII. Conclusion

This paper presented an execution-grounded hierarchical framework for multi-UAV maritime data collection. A legal rate-based candidate generator defines the high-level assignment set, an HGNN scorer ranks complete joint assignments using listwise labels derived from 15-slot execution payload, and a residual MAPPO controller executes each selected task through per-slot motion corrections and safety projection. In the evaluated 6-UAV, 12-buoy simulator, Scheme C improves mean 48-slot payload over G1_RATE $k = 6$ on both development validation and a separate holdout set. The gains are positive on average but not universal, and low-level optimization experiments remain inconclusive. The current results therefore support execution-grounded high-level ranking as a modular improvement in the tested setting, while identifying task-distribution robustness, candidate efficiency, and temporal alignment as open problems.

Reproducibility Note

The implementation is maintained in the repository https://github.com/ISAAC558345/waypoint-transformer/tree/feature/maritime-wt-stage1a. The formal configuration uses Scheme C Epoch 1, G1_RATE $k = 6$, hold $H = 15$, and the frozen MAPPO-V2 checkpoint `phase6g_seed0.pt`. The formal regression scenario 322002 is reported at approximately 69.252 Mbit, subject to the repository's bit-accounting tolerance. Before publication, the authors should archive the exact code commit, dependency lockfile, configuration hashes, and raw per-scenario benchmark outputs.

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